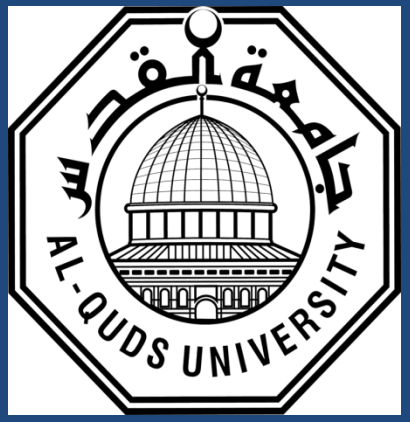


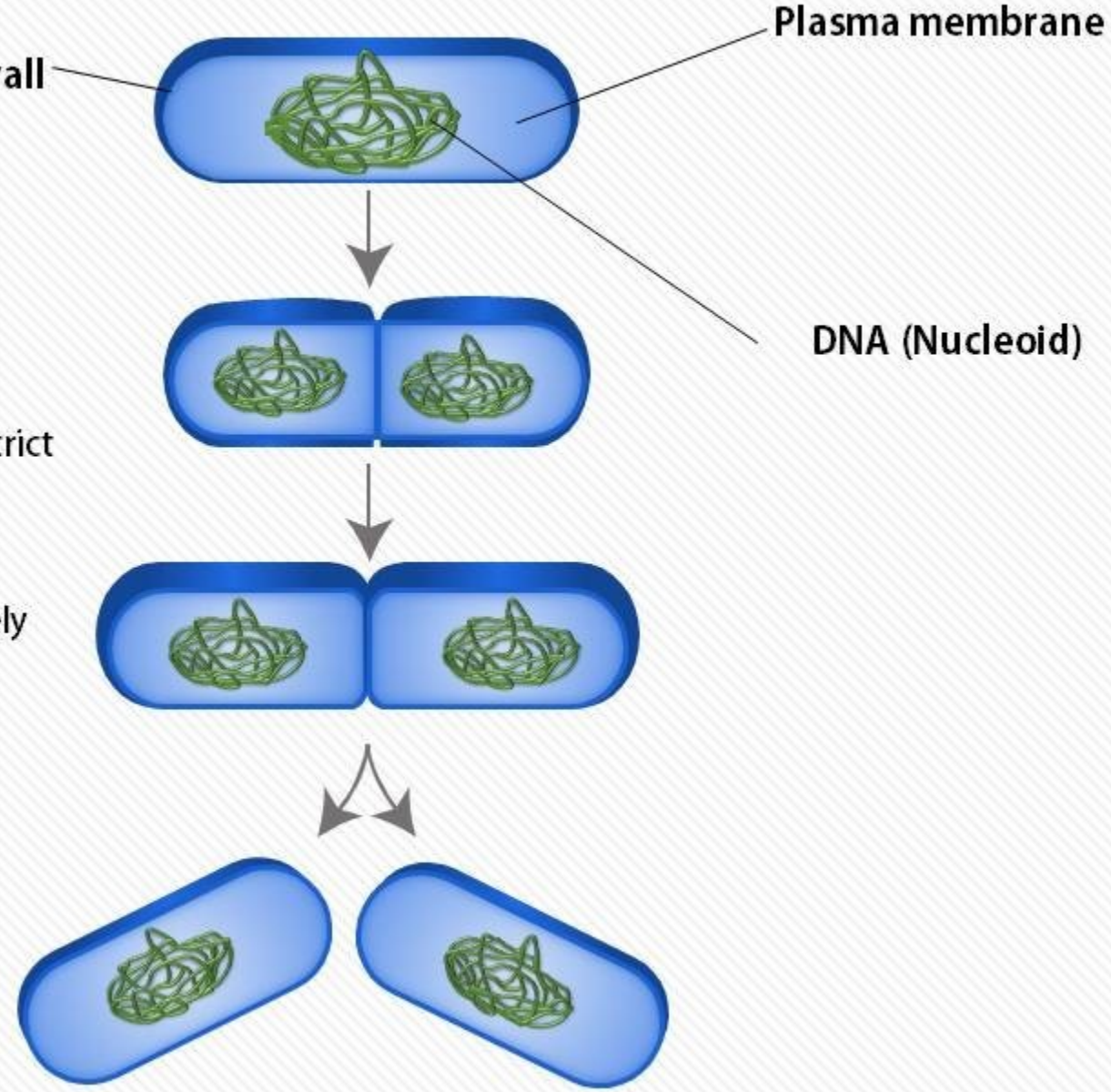
Bacterial Growth

GROWTH CYCLE

- Reproduction by binary fission
- Two identical cells are formed
- Bacteria are said to undergo exponential growth!!!

Binary Fission

1. Cell elongate DNA is replicated.
2. Cell wall and plasma membrane begin to constrict
3. Cross - wall falls completely separating the two DNA copies
4. Cell s separate



Generation Time

- Generation time or Doubling time:

the interval of time between successive binary fission of a cell or population of cells

- Generation time varied from: 20 minutes for *Escherichia coli* up to 18 hours for *Mycobacterium tuberculosis.*

- Factors that affecting doubling time: Species, nutrients, temperature, pH,,,,

Theoretical final bacterial population after 24 hours of replication: (4.72×10^{21})

4,722,366,482,869,645,213,696

in words:

Four sextillion, seven hundred twenty-two quintillion, three hundred sixty-six quadrillion, four hundred eighty-two trillion, eight hundred sixty-nine billion, six hundred forty-five million, two hundred thirteen thousand, six hundred ninety-six.

Generation Time

- Some bacteria have very short generation time, for others, it is quite long

TABLE 19.1. Generation time for some bacteria under optimum conditions

Bacterium	*Medium*	*Temp(°C)*	*Generation time (min.)*
Bacillus thermophilus	Broth	55	1.3
Escherichia coli	Broth	37	20
Bacillus subtilis	Broth	37	27
Streptococcus lactis	Milk	37	30
Lactobacillus acidophilus	Milk	37	66-87
Mycobacterium tuberculosis	Synthetic	37	792-932
Treponema pallidum	Rabbit testes	37	1980

Bacterial Growth Curve

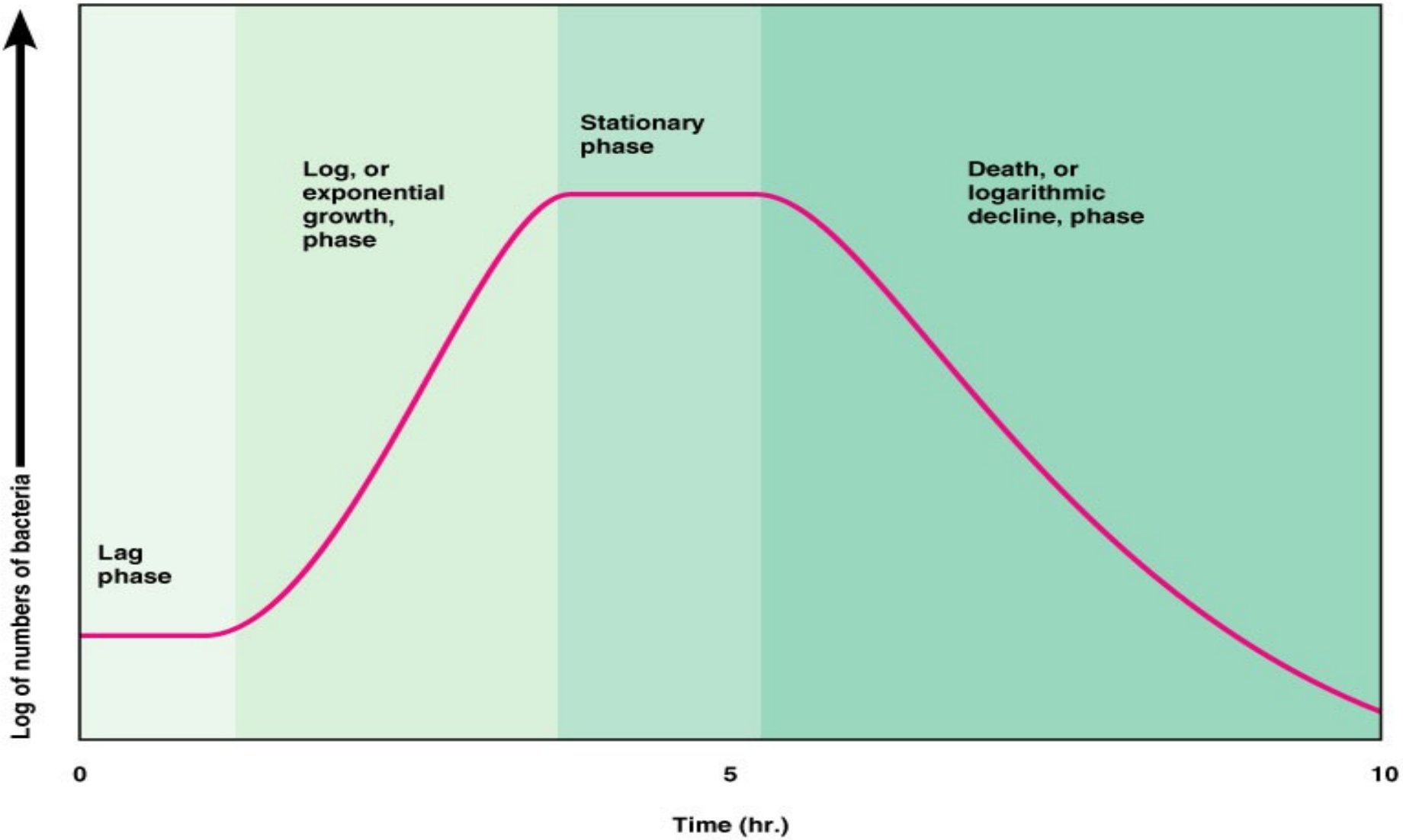
1- lag phase, during which vigorous metabolic activity occurs but cells do not divide.

2- The log / exponential phase (logarithmic) phase is when rapid cell division occurs. High metabolic rate which is the potential drug acting phase.

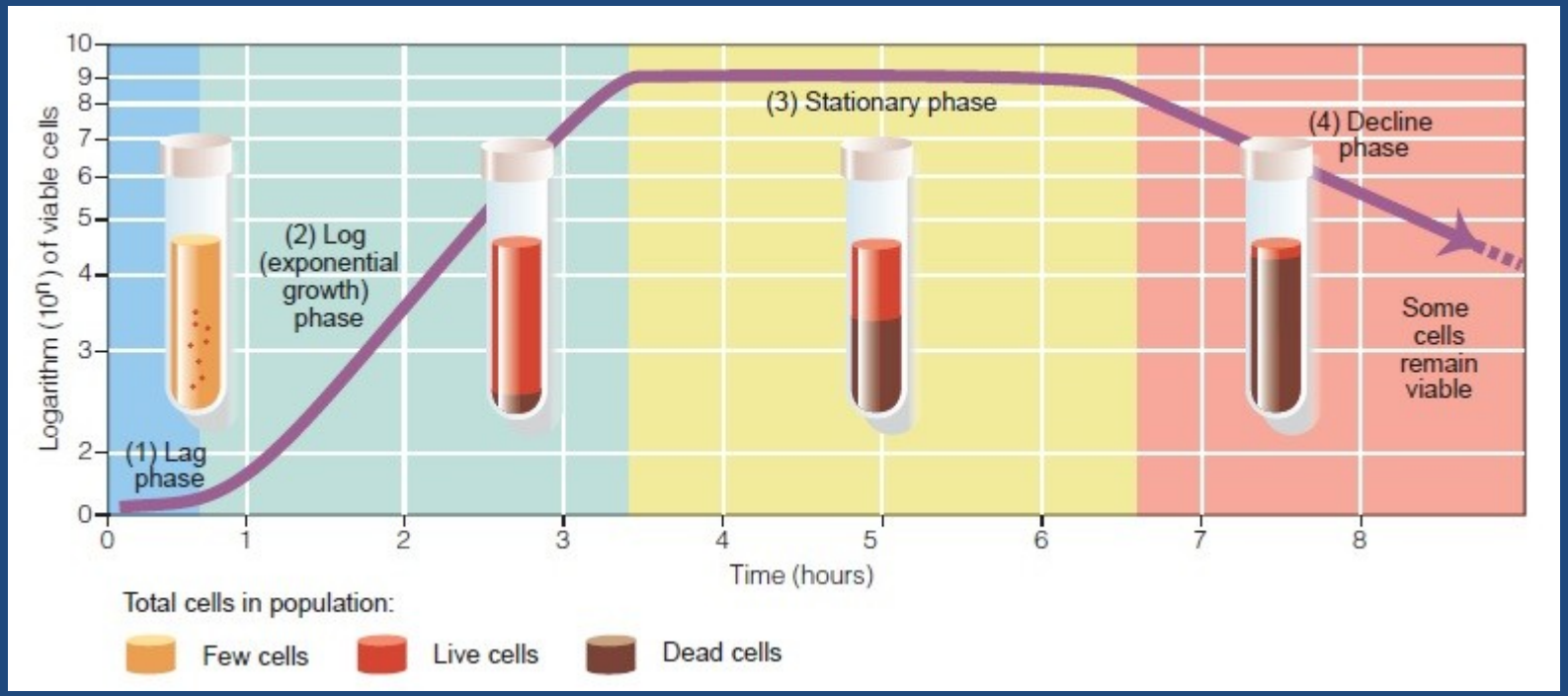
3- The stationary phase occurs when nutrient depletion or toxic products cause growth to slow.

4- The death phase, which is marked by a decline in the number of viable bacteria.

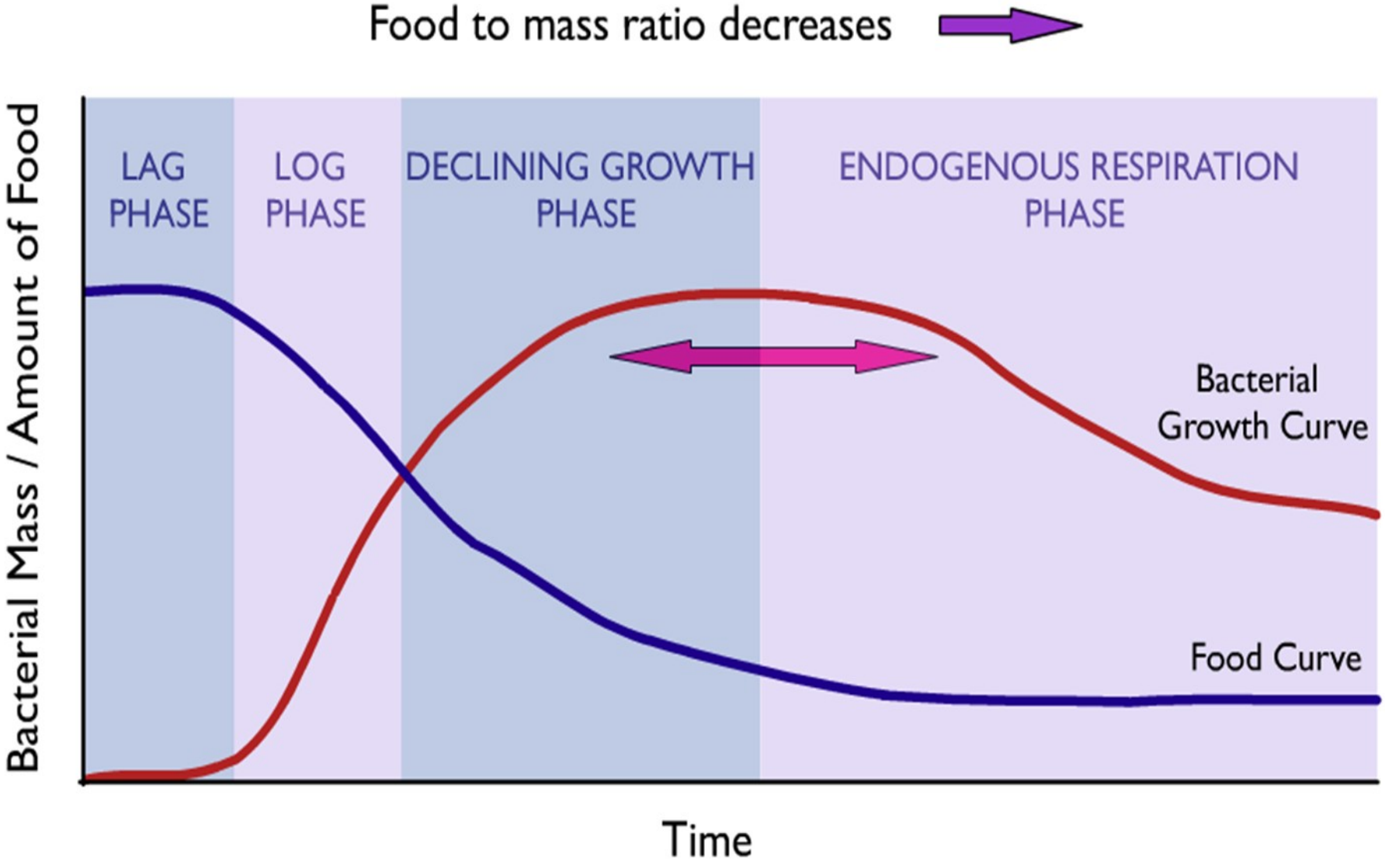
Bacterial Growth Curve



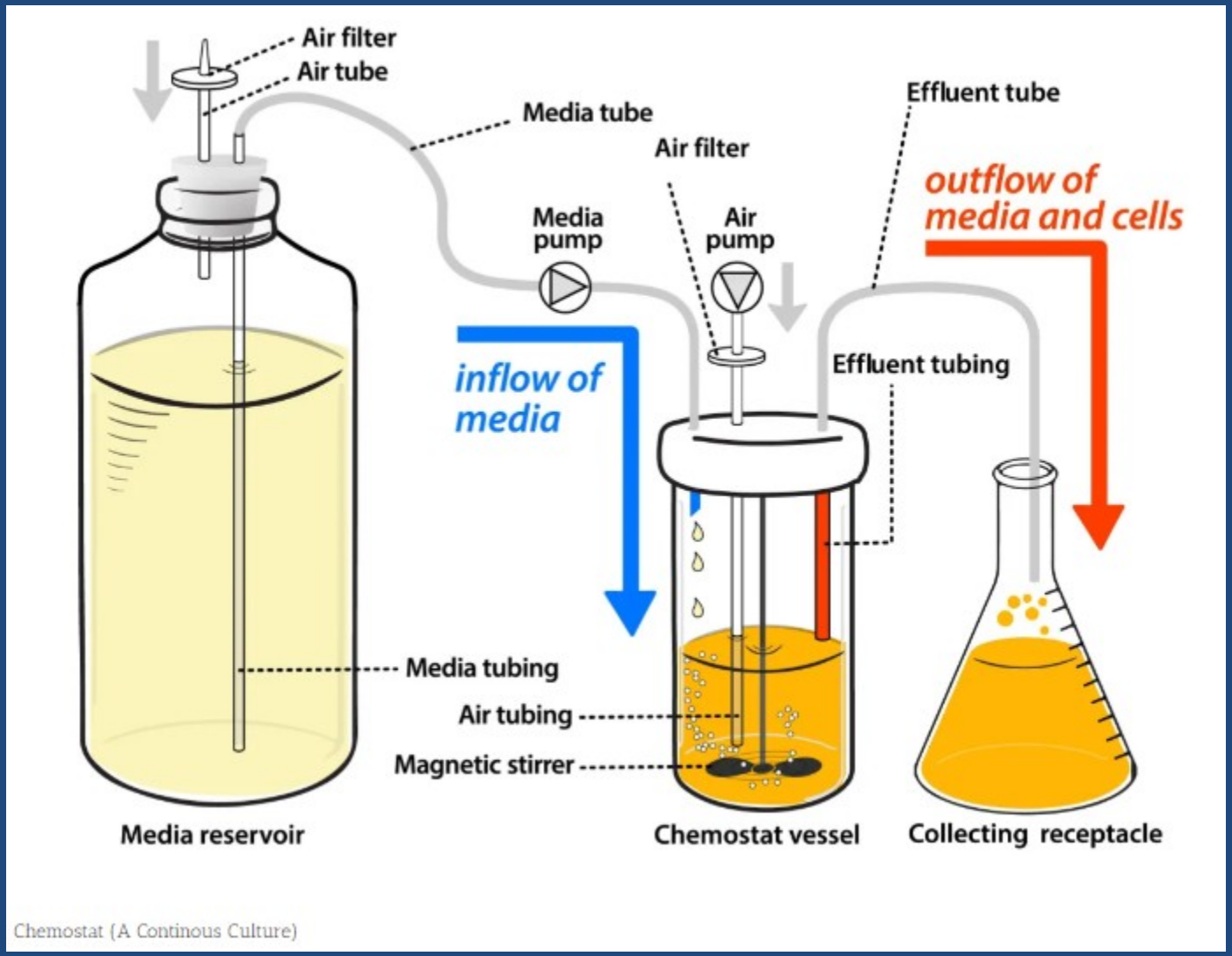
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Logarithm (10^n) of viable cells
10
9
8
7
6
5
4
3
2
0
0
1
2
3
4
5
6
7
8
Time (hours)
(1) Lag phase
(2) Log (exponential growth) phase
(3) Stationary phase
(4) Decline phase
Some cells remain viable
Total cells in population:
Few cells
Live cells
Dead cells

BACTERIAL GROWTH



Cells grown in a special apparatus called a “chemostat,” into which fresh nutrients are added and from which waste products are removed continuously, can remain in the log phase and do not enter the stationary phase.



Chemostat (A Continous Culture)

Most bacteria can grow on prepared media that include:

- Purified chemicals: sugars, amino acids, and salts.

- Blood is added for nutritional purposes

- Some bacteria (*Chlamydia*, *Rickettsia*, *Ehrlichia*, *Anaplasma)* can *only* grow within living cells.

They are obligate intracellular: it is obligatory that they grow within cells because they lack the ability to produce sufficient ATP and must use ATP that produced by the host cells.

AEROBIC & ANAEROBIC GROWTH

For most organisms, an adequate supply of oxygen enhances metabolism and growth. The oxygen acts as the hydrogen acceptor in the final steps of energy production catalyzed by the flavoproteins and cytochromes.

The use of oxygen generates two toxic molecules:

1- Hydrogen peroxide (H_2O_2)

2- Free radical superoxide (O_2^-).

bacteria require two enzymes to detoxify these molecules:

1- superoxide dismutase, which catalyzes the reaction

$2O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$

2- catalase, which catalyzes the reaction

$2H_2O_2 \rightarrow 2H_2O + O_2$

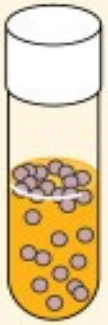
The response to oxygen is an important criterion for classifying bacteria:

1- Obligate aerobes : they require oxygen to grow (*M. tuberculosis)*.

2- Facultative anaerobes : (*E. coli*) they utilize oxygen if it is present, to generate energy by respiration, but they can use the fermentation pathway to synthesize ATP in the absence of oxygen.

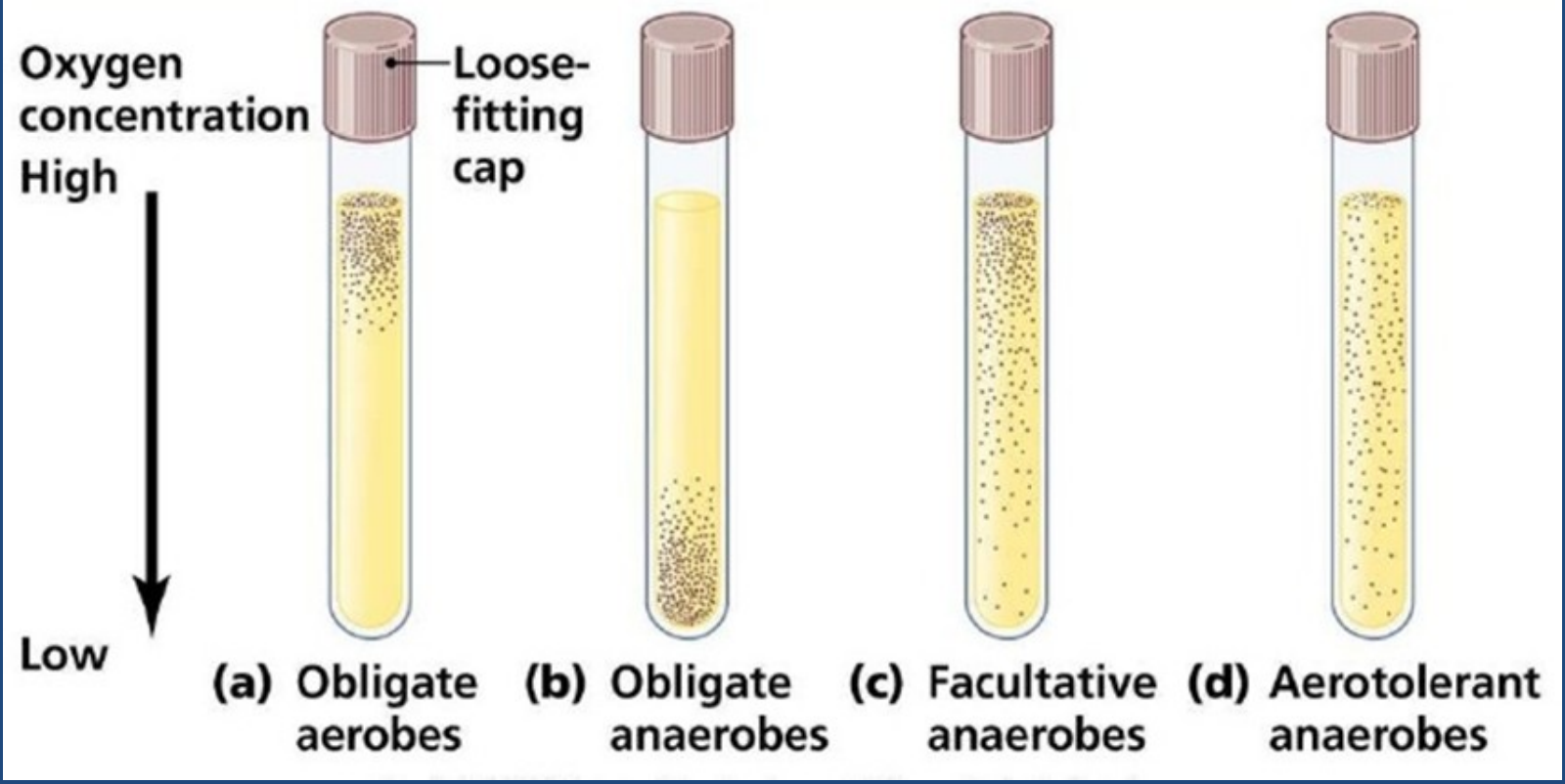
3- Obligate anaerobes: (*Clostridium tetani)*, cannot grow in the presence of oxygen, because they lack either superoxide dismutase or catalase, or both. They need oxygen in different degrees; some can survive but are not able to grow, whereas others are killed rapidly.

TABLE 6.1 **The Effect of Oxygen on the Growth of Various Types of Bacteria**

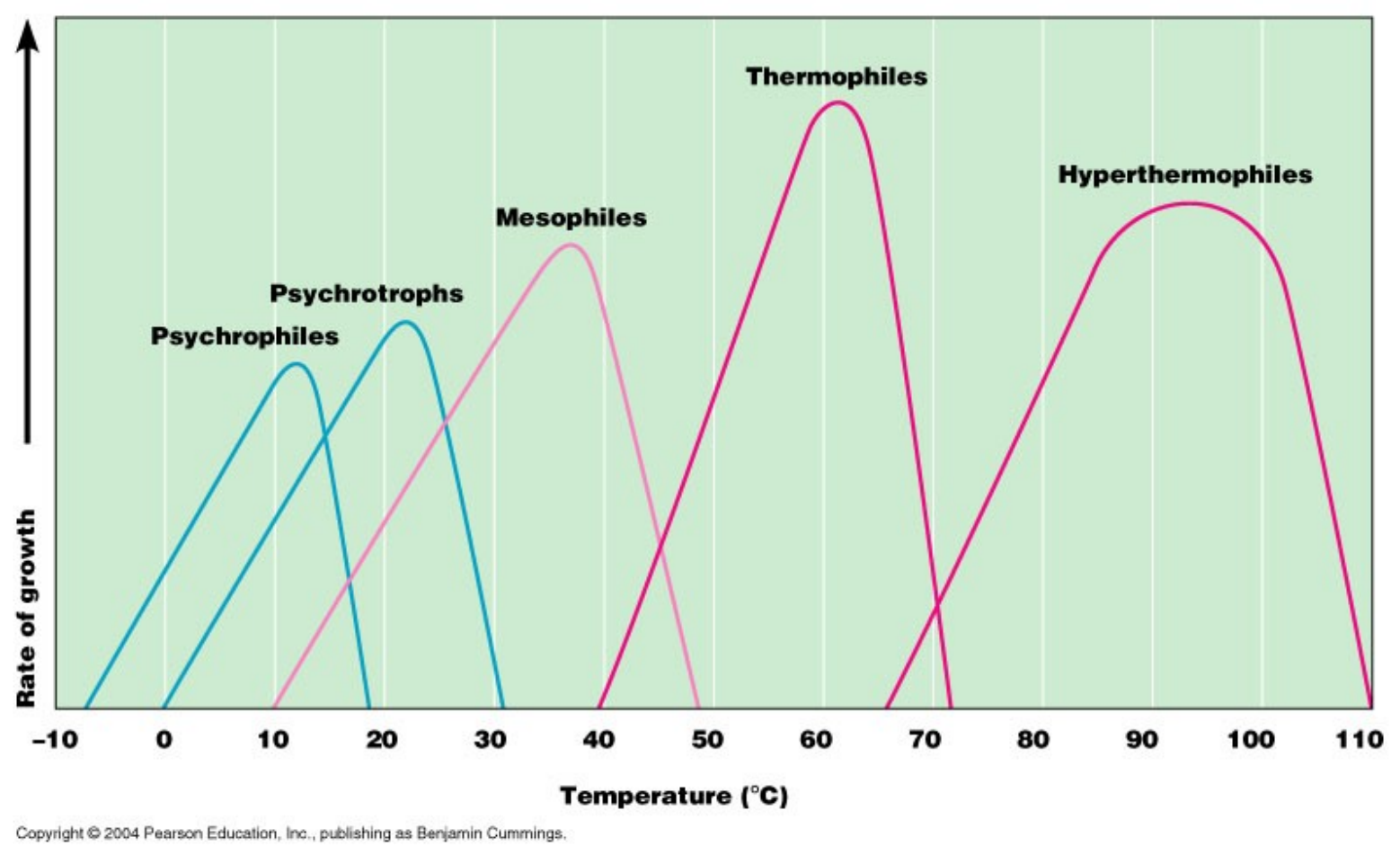
	a. Obligate Aerobes	b. Facultative Anaerobes	c. Obligate Anaerobes	d. Aerotolerant Anaerobes	e. Micro-aerophiles
Effect of Oxygen on Growth	Only aerobic growth; oxygen required.	Both aerobic and anaerobic growth; greater growth in presence of oxygen.	Only anaerobic growth; ceases in presence of oxygen.	Only anaerobic growth; but continues in presence of oxygen.	Only aerobic growth; oxygen required in low concentration.
Bacterial Growth in Tube of Solid Growth Medium					
Explanation of Growth Patterns	Growth occurs only where high concentrations of oxygen have diffused into the medium.	Growth is best where most oxygen is present, but occurs throughout tube.	Growth occurs only where there is no oxygen.	Growth occurs evenly; oxygen has no effect.	Growth occurs only where a low concentration of oxygen has diffused into medium.
Explanation of Oxygen's Effects	Presence of enzymes catalase and superoxide dismutase (SOD) allows toxic forms of oxygen to be neutralized; can use oxygen.	Presence of enzymes catalase and SOD allows toxic forms of oxygen to be neutralized; can use oxygen.	Lacks enzymes to neutralize harmful forms of oxygen; cannot tolerate oxygen.	Presence of one enzyme, SOD, allows harmful forms of oxygen to be partially neutralized; tolerates oxygen.	Produce lethal amounts of toxic forms of oxygen if exposed to normal atmospheric oxygen.

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Effects of Oxygen concentration on bacterial growth

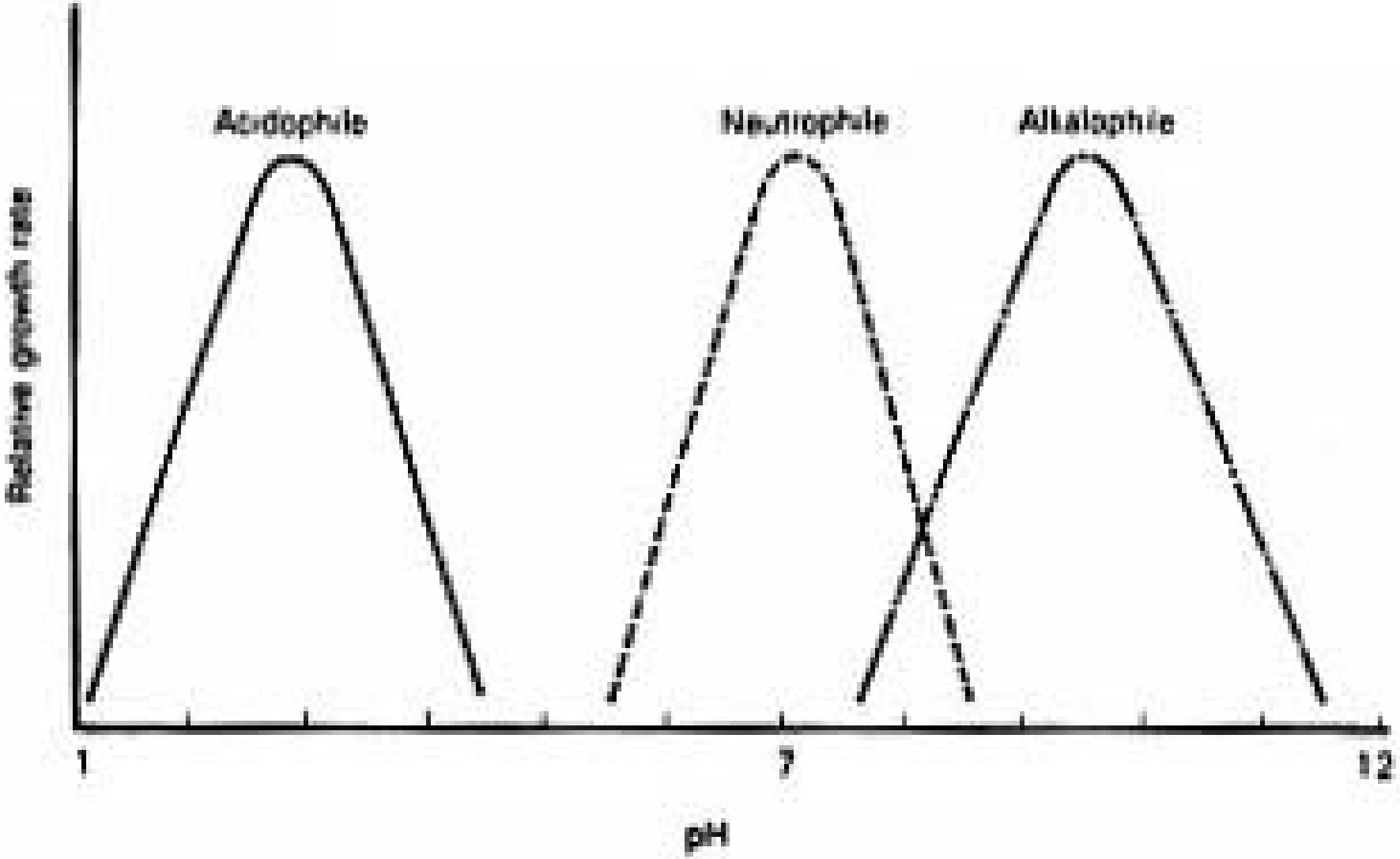


The Effect of Temperature on bacterial Growth

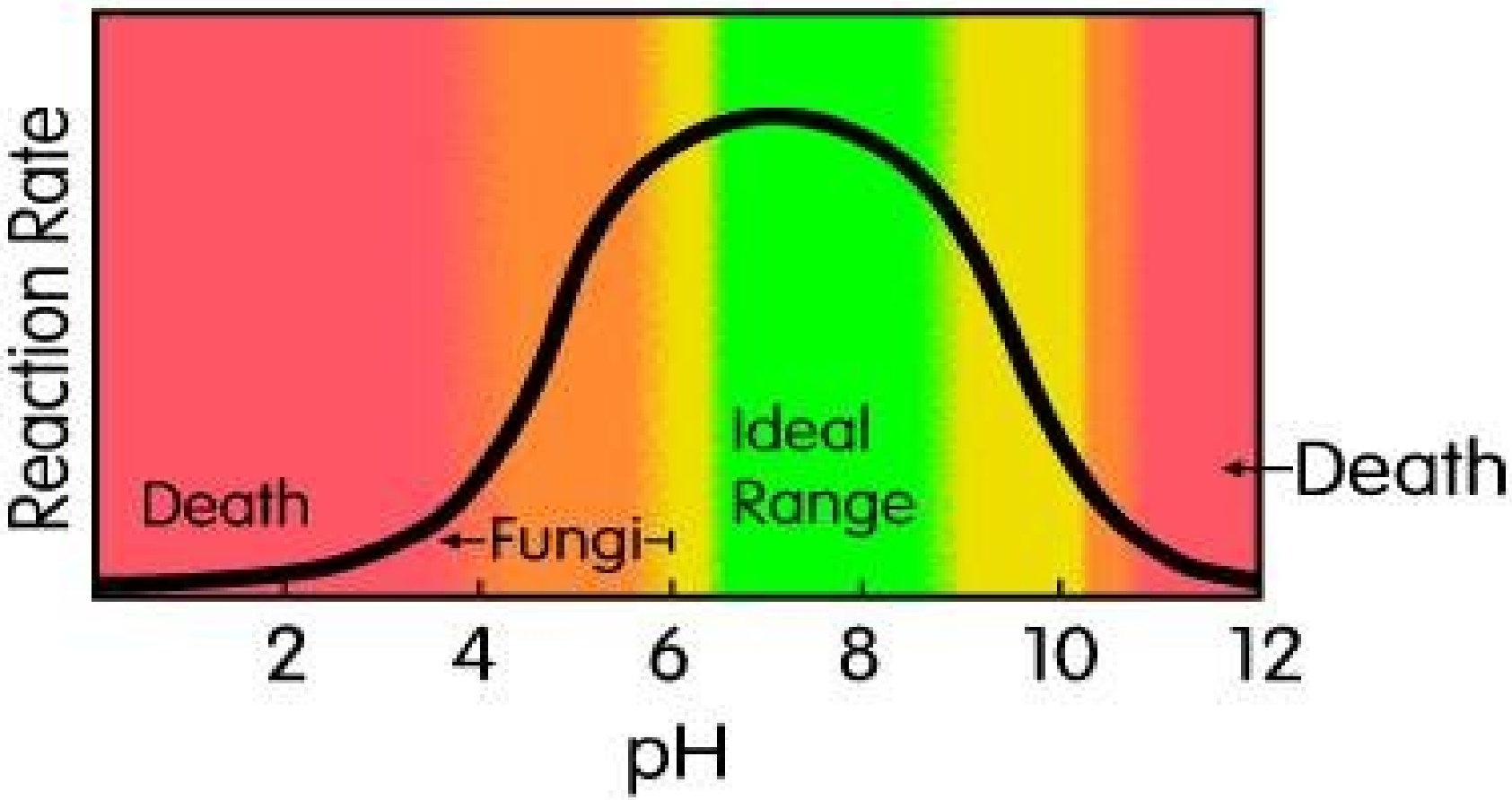


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The Effect of pH on bacterial Growth



The Effect of pH on bacterial Growth



Other terms of bacterial growth

- **Halophilic:** organisms requiring high salt concentrations

- **Osmophilic:** organisms requiring high osmotic pressure

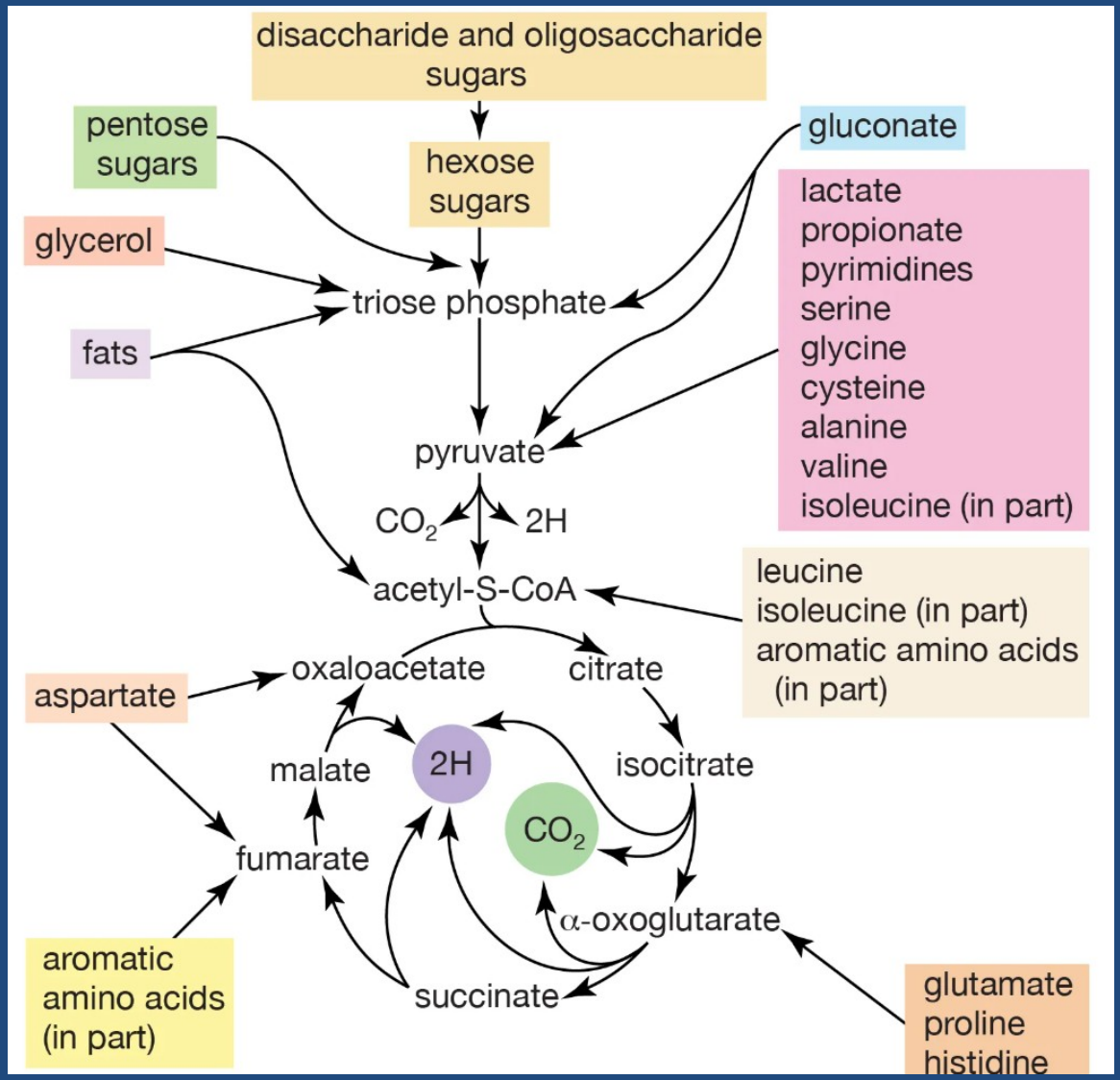
Metabolism

- **The Greek *metabole*, meaning change.**
- **Metabolism is the sum of Catabolism and Anabolism**
- **It is the totality of an organism's chemical processes to maintain life.**

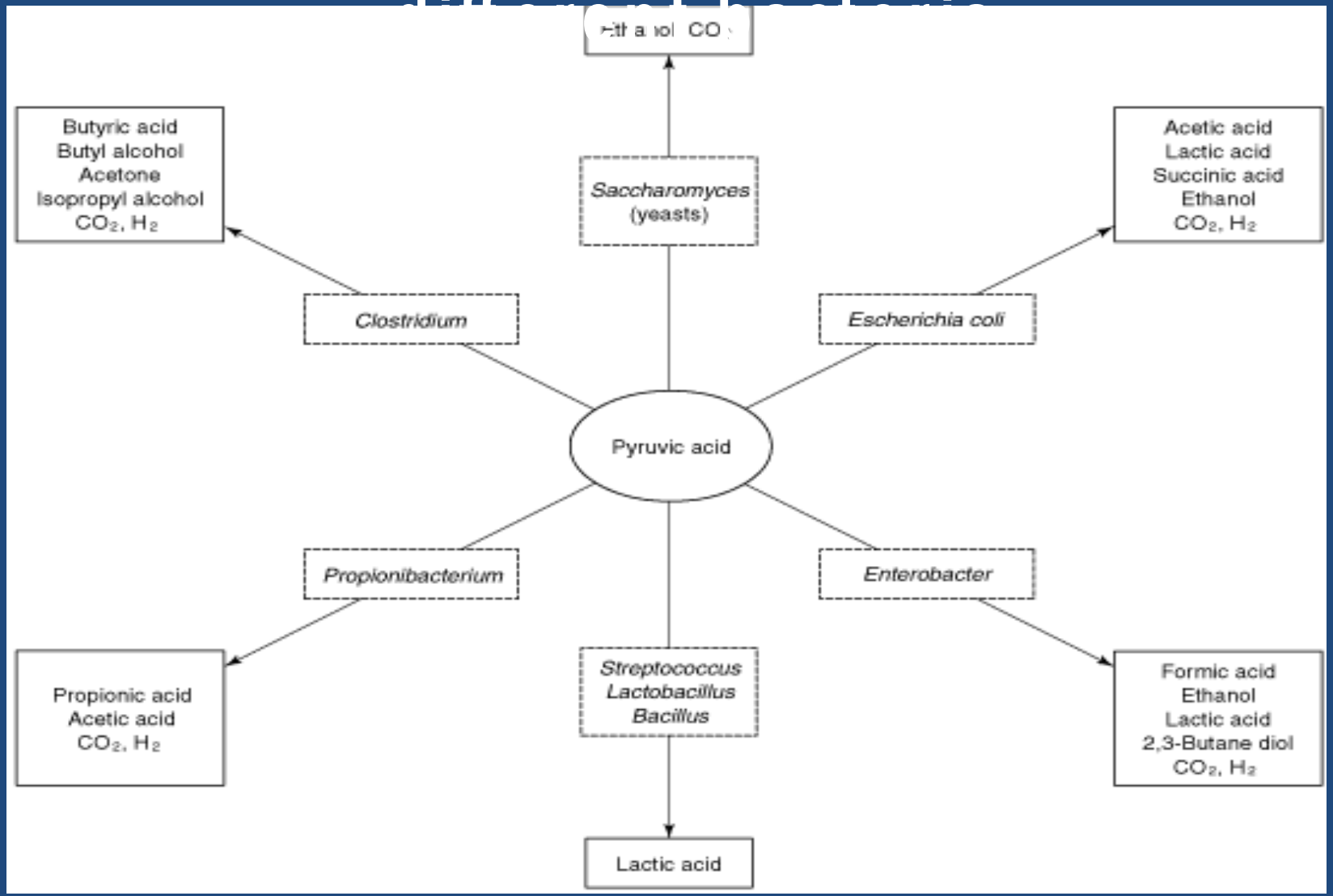
  - Catabolism

  - Anabolism

Overview of Metabolism



Overview of fermentation products formed from pyruvic acid by different bacteria



FERMENTATION OF SUGARS:

fermentation is the breakdown of a monosaccharide sugar (glucose or maltose) to pyruvic acid and then lactic acid

Fermentation is the process by which facultative bacteria generate ATP in the absence of oxygen.

If oxygen is present, the pyruvate produced by fermentation enters the Krebs cycle and is metabolized into two final products, CO_2 and H_2O.

FERMENTATION OF SUGARS:

- The Krebs cycle generates much more ATP than the glycolytic cycle; therefore, facultative bacteria grow faster in the presence of oxygen.

- Facultative and anaerobic bacteria do fermentation. but aerobes do not.

- Aerobes, such as *Pseudomonas aeruginosa*, produce metabolites that enter the Krebs cycle by processes other than fermentation, such as the deamination of amino acids.

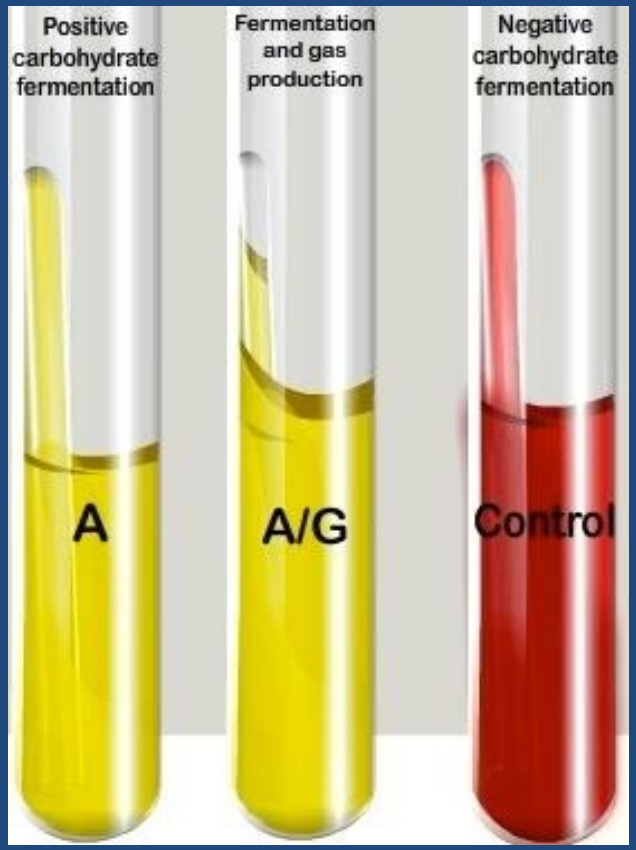
Sugar fermentation is a criteria that used in bacterial identification:

- *Neisseria gonorrhoeae* and *Neisseria meningitidis* can be distinguished from each other on the basis of fermentation of either glucose or maltose. *(N. gonorrheae utilize only glucose, but N. meningitidis produces acid from both glucose and maltose).*

- *E. coli* can be differentiated from *Salmonella* and *Shigella* on the basis of fermentation of lactose .

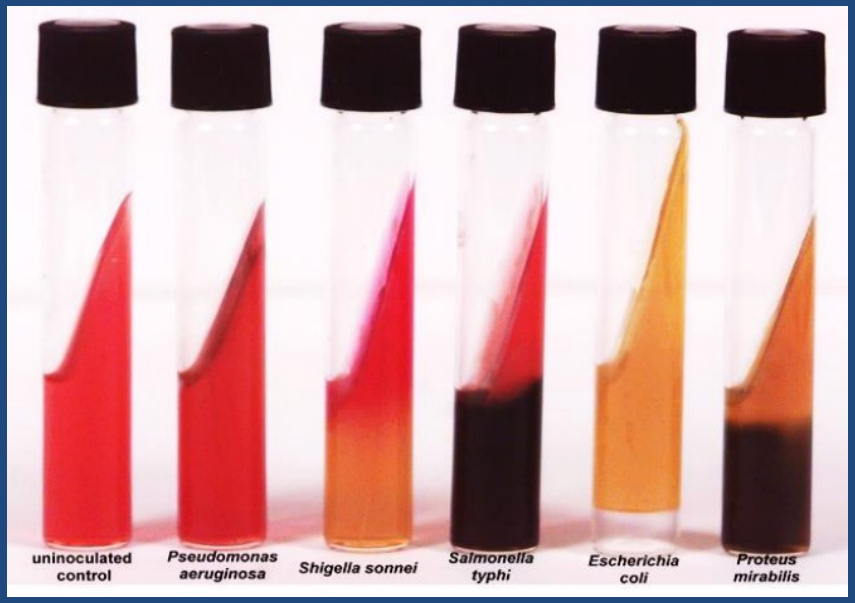
- Fermentation tests: the production of pyruvate and lactate turns the medium acid, which can be detected by (phenol red) a pH indicator.

- So, if a sugar is fermented; the medium turns yellow. If the sugar is not fermented, the phenol red remains red.


Positive carbohydrate fermentation
A
Fermentation and gas production
A/G
Negative carbohydrate fermentation
Control

Triple Sugar Iron (TSI) test





Application of sugar fermentation as a differential biochemical test.

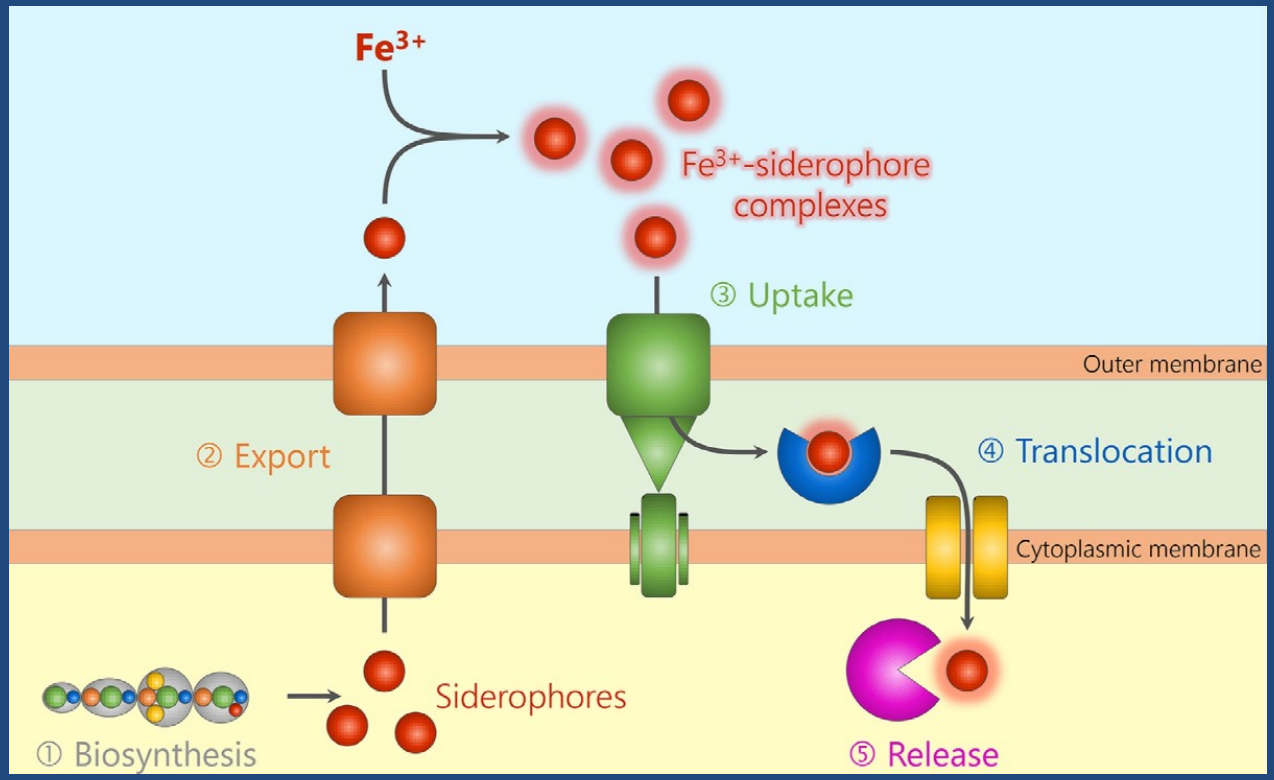
IRON METABOLISM

Ferric ion, is required for the growth of bacteria because it is an essential component of cytochromes and other enzymes.

Iron in human body is available in very low amount; because the iron is sequestered in iron-binding proteins such as transferrin.

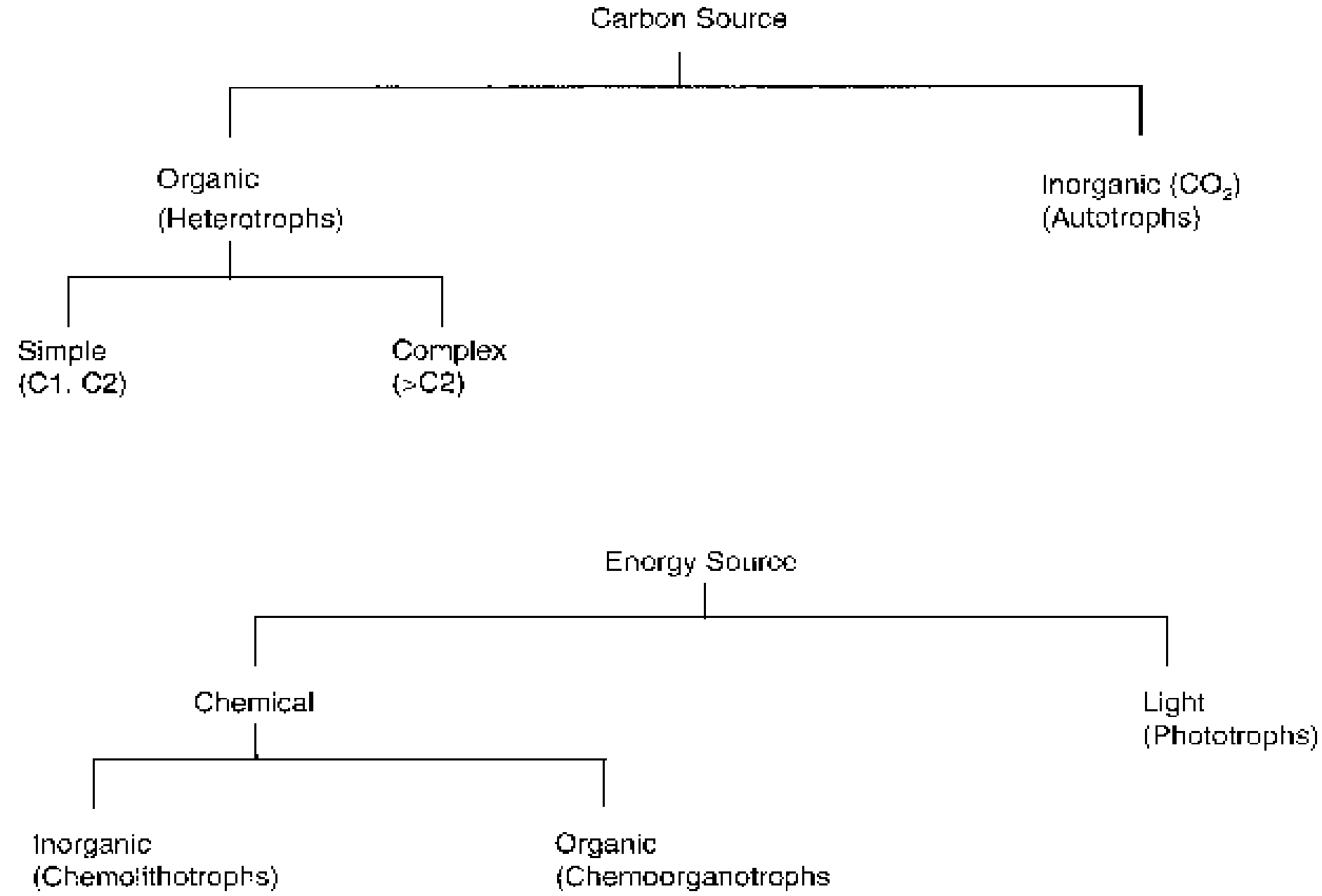
Bacteria produce iron-binding compounds called siderophores.

Siderophores, such as enterobactin produced by *E. coli*, are secreted by the bacteria, capture iron by chelating it (they compete with host's iron binding proteins), then attach to specific receptors on the bacterial surface, and are actively transported into the cell where the iron becomes available for use.



Siderophore-dependent iron uptake.
Bacteria rely on high-affinity surface receptor proteins (green) that bind iron-loaded siderophores.

Classification by Nutritional Requirement



Energy sources and biological systems

Autotrophs (self-feeding): synthesize their foods from simple carbon sources

- **Photoautotrophs** (cyanobacteria) synthesize foods using light energy and carbon dioxide gas (photosynthesis)

- **Chemoautotrophs**, carbon dioxide + use chemical reactions to obtain energy from inorganic compounds (chemosynthesis)

Energy sources and biological systems

- **Heterotrophs** (other-feeders)

1. **Photoheterotroph**, light + organic compound fatty acid or alcohol (carbon source)

2. **Chemoheterotrophs**, organic compounds for energy and carbon (glucose)

Parasite: that feed on living organic matter

Four basic groups of organisms based on their carbon and energy sources.

Carbon source	Energy source: Light *(photo-)*	Energy source: Chemical compounds *(chemo-)*
Carbon dioxide *(auto-)*	***Photoautotrophs*** • Plants, algae, and cyanobacteria use H_2O to reduce CO_2, producing O_2 as a by-product • Green sulfur bacteria and purple sulfur bacteria do not use H_2O nor produce O_2	***Chemoautotrophs*** • Hydrogen, sulfur, and nitrifying bacteria, some archaea
Organic compounds *(hetero-)*	***Photoheterotrophs*** • Green nonsulfur bacteria and purple nonsulfur bacteria, some archaea	***Chemoheterotrophs*** • Aerobic respiration: most animals, fungi, and protozoa, and many bacteria • Anaerobic respiration: some animals, protozoa, bacteria, and archaea • Fermentation: some bacteria, yeasts, and archaea